

COMP2300/6300 Mid-Semester Exam

Semester 1 2017

Name:

Student ID:

Time: 15 minutes reading, 90 minutes writing

Total marks: 100

Weighting: 13%

Permitted materials: 1 double-sided A4 cheat sheet

Instructions:

Questions are not equally weighted, and the size of the answer box is not necessarily related to the number of marks given for the question.

All answers must be written in the boxes provided in this booklet. You will be provided with scrap paper for working, but only the answers written in this booklet will be marked. Do not remove this booklet from the examination room. There is additional space at the end of the booklet in case the boxes provided are insufficient. If you use these extra pages, make sure you clearly label which question the answer refers to.

Greater marks will be awarded for short, concrete answers than long, vague/rambling ones. Marks may be deducted for providing information that is irrelevant to a question.

--	--	--	--	--

For examiner use

Question 1: Code Comprehension (10 marks)

```
.syntax unified
.global main

.macro foo bar
    ldr r1, [\bar]
    ldr r2, [\bar, #4]
    cmp r1, r2
    beq end
.endm

main:
    ldr r0, =numbers

label1:
    foo r0
    add r0, #4
    b label1

end:
    b end

.data
numbers:
    .word 2, 3, 0, 0, 6, 3, 0, 0
```

Part 1 (5 marks)

Once this program reaches the **end:** label, what values will be in **r1** and **r2**?

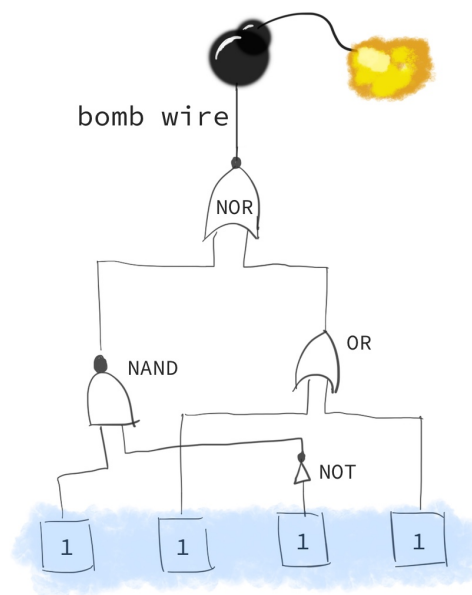
Answer:

Part 2 (5 marks)

In one English sentence, what does this ARM assembly program do?

Answer:

Question 2: Defuse the Bomb (30 marks)



You're a bomb defusal expert, and you come across this device. The four digits at the bottom represent a 4-bit binary counter which will count down from **1111** (as shown) to **0000**, with the least significant bit on the right. The bomb will explode when its "input wire" is **1**.

Part 1 (10 marks)

At what counter value will the bomb go off? Explain your answer.

Answer:

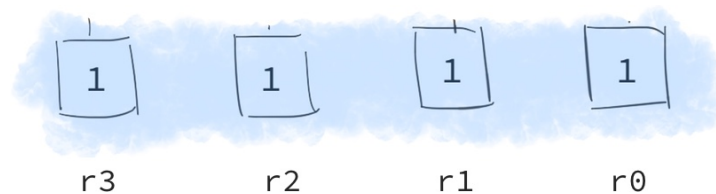
Part 2 (10 marks)

Imagine you have **one** battery which you can attach to **one** of the four counter bits to keep it at 1 regardless of the state of the counter. Is there a single counter bit you could attach it to to stop the bomb going off? Explain your answer.

Answer:

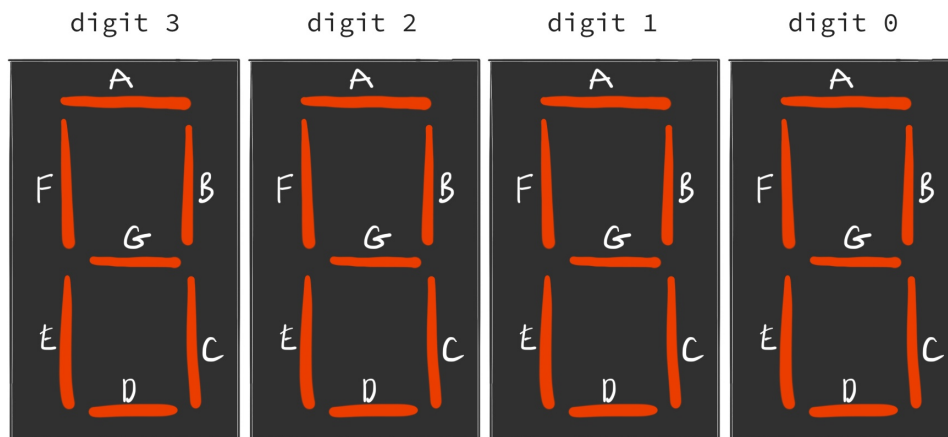
Part 3 (10 marks)

Assume the values of the four counter bits are stored in the least significant bit of the registers **r0** to **r3** as shown. Write a small ARM assembly program to calculate the value (0 or 1) of the bomb wire and store it in **r4**.



Answer:

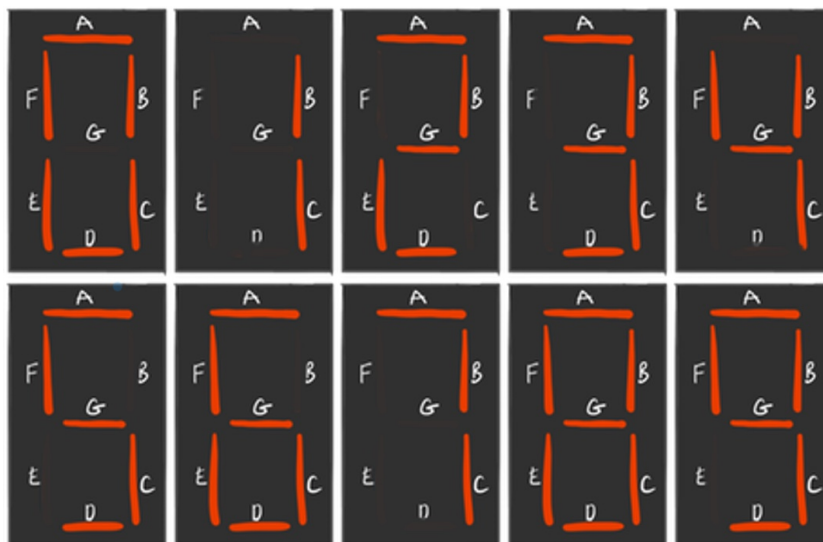
Question 3: 7-segment display (30 marks)



You've been asked to program a hardware device with a four-digit display like the one above. The device is just like your discoboard but with an additional 32-bit display register **dr** which controls the display. Each byte of the register controls the segments for a different display digit. For example, if the "A" bit for digit 2 is 1, the top segment of digit 2 will be on, if the same bit is 0, it'll be off (since there are only 7 segments, the top bit in each byte is unused).



This is what all the digits 0–9 should look like on this display.



Part 1 (10 marks)

In ARM assembly, write a short program which writes a value to the **dr** register to show the number 7019 on the display.

Answer:

Part 2 (10 marks)

Write a new ARM assembly program which takes a **single-digit** decimal number (i.e. 0–9) in the register **r0** and displays it on the right-most digit of the display (digit 0). Include labels & comments where appropriate.

Answer:

Part 3 (10 marks)

Write a more sophisticated ARM assembly program which takes a **four-digit** decimal number (i.e. 0000–9999) in the register **r0** and displays it across the four digits of the display. You may refer to any functions/macros from your answer to Part 2, or you may write a self-contained program. Include labels & comments where appropriate.

Answer:

Answer:

Question 4: Function calling conventions (30 marks)

Part 1 (20 marks)

Write an ARM assembly program which contains a function which implements Einstein's famous $E = mc^2$ equation, passing arguments & returning the result using **registers only**.

Answer:

Answer:

Part 2 (10 marks)

Re-write the $E = mc^2$ function, but this time passing arguments & returning the result using the **stack only** (i.e. no registers involved in parameter passing and returning).

Answer:

Answer:

